

Disk Granulometries on the Digital Grid: Why a Single Structuring Element Is Not Enough

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Abstract. The usual way to measure size distributions is to open an image by the digital disks of increasing radius that common libraries provide. We show that this is not a granulometry: these disks violate Matheron's absorption axiom for most pairs of radii, and the resulting size distributions can increase with the radius. Two remedies exist. Replacing each opening by the union of openings by all larger disks yields a correct granulometry that is as round as the disks themselves and costs the same; we recommend it. Keeping a *single* structuring element per size, as in octagons, periodic lines or Bresenham decompositions, forces polygonal shapes, and we prove that this is unavoidable. If the mean radius of a Minkowski chain grows by at most δ per element, every element of radius ρ deviates from every disk by at least about $\rho/(16\pi^2\delta^2)$, in any dimension. An explicit chain of periodic lines attains this rate within a factor of 4, and for a fixed library the exact slope is attained by octagons. Experiments on synthetic and real images confirm the diagnosis, the remedy, and when the anisotropy of single-element families matters.

Keywords: Granulometry · Digital disk · Minkowski sum · Lattice polygon · Width function · Pattern spectrum

1 Introduction

Granulometries, introduced by Matheron [10], formalise sieving: a family of openings $(\gamma_\lambda)_{\lambda \geq 0}$ that are decreasing in λ and satisfy the absorption law $\gamma_\lambda \gamma_\mu = \gamma_\mu \gamma_\lambda = \gamma_{\max(\lambda, \mu)}$. The size distribution and pattern spectrum derived from them [9, 14, 15] are among the most widely used morphological descriptors. The textbook construction uses a *single structuring element per size*: $\gamma_\lambda(X) = X \circ B_\lambda$. Absorption then holds if and only if B_μ is B_λ -open, i.e. a union of translates of B_λ , for all $\lambda \leq \mu$. Euclidean disks satisfy this condition.

On the grid $\mathbb{Z}^2$ they do not. The digital disks $\{p \in \mathbb{Z}^2 : \|p\| \leq r\}$ of common software are not open with respect to each other (Sect. 3): with `scikit-image` [19], 73% of the pairs of radii up to 40 violate the condition. Size distributions computed as $r \mapsto |X \circ D_r|$, the most direct use of these libraries, then increase with the radius on most random test images. There are two ways out.

- **Several structuring elements per size.** The sup-closure $\gamma'_r = \bigvee_{s \geq r} \gamma_{D_s}$, i.e. the union of the openings by all disks of radius at least r , is a granulometry for any family (D_s) (Proposition 1). With digital disks it is correct, as

round as the disks, and computable with one opening per radius. Opening transforms [18] and local thickness [5] implicitly rely on this construction. We recommend it in place of single disk openings.

- **One structuring element per size.** The classical alternative builds the elements as *Minkowski chains* $D_k = D_{k-1} \oplus C_k$, which are automatically granulometric [1,8,15,18]. Examples are octagons from neighbourhood sequences [12,4], periodic lines [8] and Bresenham segments [1]. These families are polygonal. We prove that this is unavoidable and determine the price.

Our contributions are:

- **Diagnosis** (Sect. 3). We quantify the violation of the axiom by standard digital disks and its effect on size distributions.
- **Lower bound for single-element chains** (Theorem 1, Corollaries 1–3). If the mean radius of a Minkowski chain grows by at most δ per element, then *every* element of radius ρ has error $\varepsilon \geq (1 - o(1)) \rho / (16\pi^2 \delta^2)$, where the error is the Hausdorff distance to the nearest Euclidean disk. If the fitted radius grows by bounded steps, the maximal error is at least $c\rho^{1/3}$. For a fixed library the slope $\kappa/(1 + \kappa)$ is given by a linear program, and octagons attain it.
- **Matching upper bound and higher dimensions** (Theorems 2 and 3). An explicit chain of periodic lines achieves $\Theta(\rho/\delta^2)$, within a factor of 4 asymptotically, and $\Theta(\sqrt{\rho})$ when $\delta \sim \rho^{1/4}$. In $\mathbb{Z}^n$, $\varepsilon \geq c_n \rho \delta^{-2n}$.
- **Experiments** (Sect. 6). On synthetic and real images we compare the sup-closure, Gauss disks and single-element families, both on whole size distributions and on per-object sieve sizes.

In short, round and correct disk granulometries on $\mathbb{Z}^2$ exist, but they need several structuring elements per size. Several lines of related work bear on this problem.

- Chamfer norms [2,17] approximate Euclidean balls by polygons with a fixed set of edge directions; when their balls form Minkowski chains, as for 3×3 masks, Corollary 3 applies.
- Neighbourhood sequences [4] and zonohedra [7] optimise the roundness of a fixed library.
- Soille and Talbot [16] noted that radial decompositions by Bresenham lines do not generate granulometries.
- Normand [11] builds convex structuring elements by unions of translates instead of Minkowski sums, a construction outside the scope of our bound.

To our knowledge, no lower bound of this kind has been published.

2 Preliminaries

For a finite set $X \subset \mathbb{Z}^n$ and a unit vector u , let $h_X(u) = \max_{p \in X} \langle p, u \rangle$ be the support function and $w_X(u) = h_X(u) + h_X(-u)$ the *width function*, i.e. the width of $\text{conv } X$ in direction u . Both are additive under Minkowski addition [13]:

$$h_{X \oplus Y} = h_X + h_Y, \quad w_{X \oplus Y} = w_X + w_Y. \quad (1)$$

The *anisotropy* of X is $A(X) = \max_u w_X - \min_u w_X$, and its *mean radius* is $\bar{r}(X) = \frac{1}{2} \int w_X d\sigma$, where σ is the uniform probability measure on the sphere; $\bar{r}$ is additive as well. In the plane, Cauchy's formula gives a concrete meaning: $\bar{r}(X)$ is the perimeter of $\text{conv } X$ divided by 2π , i.e. the radius of the circle with the same perimeter. For a segment of length L it equals L/π . We say that X is ε -round with radius ρ if $d_H(\text{conv } X, B(c, \rho)) \leq \varepsilon$ for some centre c . Then $|h_X(u) - \langle c, u \rangle - \rho| \leq \varepsilon$, and hence

$$2\rho - 2\varepsilon \leq w_X \leq 2\rho + 2\varepsilon, \quad A(X) \leq 4\varepsilon, \quad |\bar{r}(X) - \rho| \leq \varepsilon. \quad (2)$$

The opening of Y by a structuring element D is $Y \circ D = \bigcup \{D+t : D+t \subseteq Y\}$. A family (D_k) generates a *single-element granulometry* $\gamma_k(Y) = Y \circ D_k$ if and only if $D_l \circ D_k = D_l$ for all $k \leq l$ [10,14]; since the relation is transitive, consecutive elements suffice. A *Minkowski chain* is a family with $D_k = D_{k-1} \oplus C_k$, and every Minkowski chain generates a single-element granulometry. Its *increments* are the C_k , its *steps* are $\bar{r}(D_k) - \bar{r}(D_{k-1}) = \bar{r}(C_k)$, and, if all C_k belong to a finite set $\mathcal{L}$ (up to translation), $\mathcal{L}$ is its *library*. Without the single-element restriction, any family of structuring elements can be turned into a granulometry.

Proposition 1 (sup-closure). *For any finite sets $D_1, \dots, D_R \subset \mathbb{Z}^n$, the operators $\gamma'_r = \bigvee_{s=r}^R \gamma_{D_s}$, $r = 1, \dots, R$, where $\gamma_D(Y) = Y \circ D$, form a granulometry.*

Proof. A supremum of openings is an opening [14], and γ'_r is decreasing in r . Let $r \leq s$. Since γ'_r and γ'_s are anti-extensive and increasing, $\gamma'_r \gamma'_s \leq \gamma'_s$ and $\gamma'_s \gamma'_r \leq \gamma'_s$. Conversely, for every $t \geq s$ we have $\gamma'_r \gamma'_s \geq \gamma_{D_t} \gamma_{D_t} = \gamma_{D_t}$ and $\gamma'_s \gamma'_r \geq \gamma_{D_t} \gamma_{D_t} = \gamma_{D_t}$, and taking the supremum over $t \geq s$ gives $\geq \gamma'_s$. $\square$

Computing γ'_r from $r = R$ down to 1 needs one opening per radius, the same cost as the single-element family γ_{D_r} . The structuring elements are unchanged, so with digital disks the sup-closure is exactly as round as the disks. The rest of the paper explains why the single-element route cannot achieve this.

3 Digital Disks Are Not Granulometric

Figure 1(a) shows the smallest instance of the problem. We tested the condition “ D_r is D_s -open” for all pairs $1 \leq s < r \leq 40$, for the digital disks of `scikit-image` [19] (disk, the Gauss digitisation $\{p : \|p\| \leq r\}$) and of OpenCV [3] (MORPH_ELLIPSE). As Table 1 shows, most pairs violate it, including most consecutive pairs. When D_r is not D_s -open, $X \circ D_r \not\subseteq X \circ D_s$ for some X , and the size distribution $r \mapsto |X \circ D_r|$ can *increase*, so the pattern spectrum takes negative values. On random 120×120 images made of 3 to 11 disks, rectangles or digital disks ($r \leq 15$), this happens for 63% (scikit-image) and 60% (OpenCV) of the images. For example, in one image the opened area grows from 2053 to 2089 pixels between $r = 8$ and $r = 9$, and `scipy.ndimage`, `scikit-image` and OpenCV agree up to border conventions.

One could try to extract a granulometric *subsequence* of digital disks. Over all origin-centred disks $\{p : \|p\|^2 \leq n\}$ with $n \leq 900$, the largest radius reachable

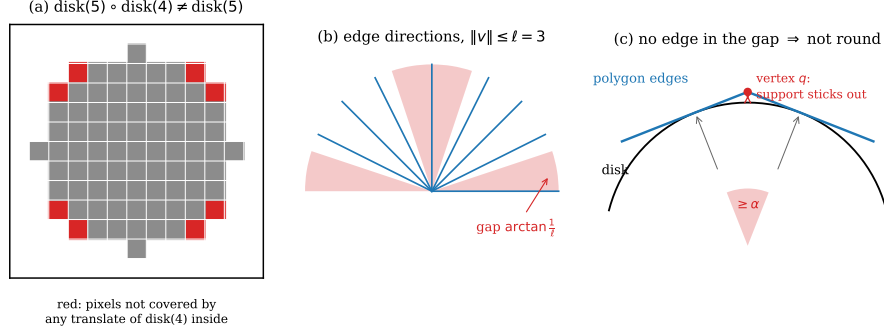


Fig. 1. (a) The digital disk of radius 5 is not open with respect to that of radius 4: the red pixels lie in no translate of the smaller disk that fits inside. (b) Idea of Theorem 1: lattice vectors of norm at most ℓ (here 3) leave gaps of $\arctan(1/\ell)$ next to the axes. (c) A polygon whose edges come from such vectors has no edge with a direction in the gap. Over the corresponding range of normals a single vertex carries the support and sticks out of the circle (schematic, gap exaggerated).

Table 1. Violations of the absorption condition by standard digital disks, $1 \leq s < r \leq 40$. The last row is the sup-closure of `disk`, which uses several elements per size and is always a granulometry; its pairwise conditions do not apply.

family	violating pairs (s, r)	violating pairs $(r - 1, r)$	images with non-monotone size distr.
<code>scikit-image disk</code>	571/780 (73%)	30/39	253/400
<code>OpenCV MORPH_ELLIPSE</code>	628/780 (81%)	36/39	241/400
sup-closure of <code>disk</code> (Prop. 1)	not applicable		0/400

by a chain of consecutively open disks with radius gaps at most G is 2.0 for $G = 1$, 9.9 for $G = 2$ and 21.8 for $G = 3$, and radius 30 is reached only with $G \geq 5$. Digital disks thus give at best very coarse granulometries.

4 Lower Bounds for Single-Element Chains

Additivity (1) turns the roundness of two elements of a chain into a constraint on the increments between them. All edges of the increments are lattice vectors. If the increments are short, these vectors leave a gap of directions around each axis (Fig. 1b), and a polygon whose edges avoid an angular interval cannot be round (Fig. 1c). Adding more elements to the chain makes the polygon larger but does not fill the gap, so the deviation grows with the radius.

Lemma 1 (direction gap). *Let $\ell \geq 1$. No nonzero $(a, b) \in \mathbb{Z}^2$ with $a^2 + b^2 \leq \ell^2$ has a direction angle in $(0, \arctan(1/\ell))$.*

Proof. A direction in that interval requires $a > 0$ and $b \geq 1$, hence $b/a \geq 1/a \geq 1/\ell$. $\square$

Theorem 1. *Let $(D_k)_{k=0}^K$ be a Minkowski chain in $\mathbb{Z}^2$ whose increments have diameter at most $\ell \geq 1$. If D_0 is ε_0 -round with radius ρ_0 and D_K is ε_K -round with radius ρ_K , then, with $c = 1 - \cos(\frac{1}{2} \arctan \frac{1}{\ell})$,*

$$4(\varepsilon_0 + \varepsilon_K) \geq 2(\rho_K - \rho_0 - \varepsilon_0 - \varepsilon_K)c. \quad (3)$$

Proof. Step 1. Every edge of $\text{conv } C_k$ joins two lattice points, so it is a lattice vector of norm at most ℓ . *Step 2.* Let $P = \sum_k \text{conv } C_k$ and $Q = \frac{1}{2}(P - P)$. By (1), $W := w_{D_K} - w_{D_0} = w_P = 2h_Q$. The edges of the centrally symmetric polygon Q are parallel to edges of the $\text{conv } C_k$, so by Lemma 1 (applied to $\pm e$) none has a direction in $(0, \alpha)$, where $\alpha = \arctan(1/\ell)$. The normal angles in $(\frac{\pi}{2}, \frac{\pi}{2} + \alpha)$ therefore belong to the normal cone of a single vertex q of Q : $h_Q(\theta) = |q| \cos(\theta - \theta_q)$ on an interval I of length at least α . *Step 3.* We may assume $s := \rho_K - \rho_0 - \varepsilon_0 - \varepsilon_K > 0$, since otherwise (3) is trivial. By (2), $\min W \geq 2s$, so $h_Q > 0$ and $|\theta - \theta_q| < \frac{\pi}{2}$ on I . On an interval of length α within $(\theta_q - \frac{\pi}{2}, \theta_q + \frac{\pi}{2})$, the function $\cos(\theta - \theta_q)$ varies by at least $1 - \cos(\alpha/2)$: by at least that much if $\theta_q \in I$, and by at least $1 - \cos \alpha$ otherwise, by monotonicity. With $|q| \geq \min h_Q \geq s$ this gives $\max W - \min W \geq 2sc$. *Step 4.* Finally $\max W - \min W \leq A(D_K) + A(D_0) \leq 4\varepsilon_K + 4\varepsilon_0$ by (2). $\square$

Only the first and the last element enter the bound. For a chain starting from a single point ($\varepsilon_0 = \rho_0 = 0$), (3) gives $\varepsilon_K \geq \rho_K c/(2 + c)$, where $c = \frac{1}{8\ell^2}(1 + O(\ell^{-2}))$: every element is at least $(1 - o(1))\rho/(16\ell^2)$ away from a disk. Different constraints on the steps bound ℓ in different ways.

Corollary 1 (mean-radius steps). *If a Minkowski chain starting from a point has mean-radius steps $\bar{r}(C_k) \leq \delta$, then $\ell \leq \pi\delta$, and every element of radius ρ has error $\varepsilon \geq \rho c/(2 + c) = (1 - o(1))\rho/(16\pi^2\delta^2)$, with $c = 1 - \cos(\frac{1}{2} \arctan \frac{1}{\pi\delta})$. In particular, granulometric chains with bounded steps have error growing linearly with the radius.*

Proof. By Cauchy's formula the perimeter of a planar convex set is π times its mean width, and it is at least twice its diameter. Hence $\text{diam conv } C_k \leq \frac{\pi}{2} \cdot 2\bar{r}(C_k) \leq \pi\delta$. $\square$

Corollary 2 (fitted-radius steps). *If all elements of a Minkowski chain are ε -round with radii satisfying $\rho_k - \rho_{k-1} \leq \Delta$, then $\varepsilon \geq (1 - o(1))((\rho_K - \rho_0)/512)^{1/3}$ as $\rho_K \rightarrow \infty$.*

Proof. By (1) and (2), $w_{C_k} = w_{D_k} - w_{D_{k-1}} \leq 2\Delta + 4\varepsilon$, so $\ell \leq 2\Delta + 4\varepsilon$. If ε stayed bounded, then so would ℓ , and (3) would force $\varepsilon \rightarrow \infty$. Hence $\ell = 4\varepsilon(1 + o(1))$, and (3) with $\varepsilon_0 = \varepsilon_K = \varepsilon$ becomes $8\varepsilon \geq (\rho_K - \rho_0)/(64\varepsilon^2)(1 + o(1))$. $\square$

Corollary 3 (fixed library). *Let $\mathcal{L}$ be a finite library containing a set with more than one point, and let*

$$\kappa(\mathcal{L}) = \inf \left\{ \frac{1}{4} A(\sum_C \lambda_C w_C) : \lambda \geq 0, \min_u \sum_C \lambda_C w_C(u) = 2 \right\}.$$

Then $\kappa := \kappa(\mathcal{L}) > 0$, and the last element of every chain with library $\mathcal{L}$ satisfies $\varepsilon_K \geq (\kappa(\rho_K - \rho_0 - \varepsilon_0) - \frac{1}{4}A(D_0))/(1 + \kappa)$, i.e. $\varepsilon \geq \frac{\kappa}{1+\kappa}\rho - O(1)$.

Proof. For λ in the simplex, $f(\lambda) = A(\sum \lambda_C w_C) / \min \sum \lambda_C w_C$ takes values in $(0, \infty]$: $\sum \lambda_C w_C$ is the width function of the polygon $\sum \lambda_C \text{conv } C$, which does not have constant width, because bodies of constant width are strictly convex [13]. Moreover f is continuous where finite and tends to ∞ where the minimum vanishes, so it attains a positive minimum, and $\kappa = \frac{1}{2} \min f > 0$. For the chain, write $W = w_{D_K} - w_{D_0} = \sum_C n_C w_C$ and $s = \frac{1}{2} \min W \geq \rho_K - \rho_0 - \varepsilon_0 - \varepsilon_K$. Then $A(W) \geq 4\kappa s$ and $A(W) \leq 4\varepsilon_K + A(D_0)$. $\square$

We compute κ by linear programming on a grid of 3600 directions, with all D_4 -images included. It is 0.2071 for $\{N_4\}$ or $\{N_8\}$ (squares) and 0.0412 for octagons. For libraries of the 8-connected digital straight segments (DSS) of all primitive v with $\|v\| \leq m$, it is 0.0176 for $m = \sqrt{5}$ and 0.0028 for $m = 10$. For the corresponding periodic lines $\{0, v\}$ it is 0.0137 and, for $m = 5$, 0.0038. Since the grid underestimates the anisotropy, the computed values are lower approximations, and the bounds remain valid. For example, $\kappa(\{N_4\}) = (\sqrt{2} - 1)/2$ is the anisotropy of the square. Corollary 3 is tight for octagons: the element of fitted radius 961.5 of the best octagon chain has $\varepsilon/\rho = 0.039571$, while $\kappa/(1 + \kappa) = 0.039570$.

5 Matching Upper Bound and Higher Dimensions

The lower bound is attained, up to constants, by a simple explicit chain. For $m \geq 1$, let V_m contain one vector from each pair $\pm v$ of integer points on the boundary of $[-m, m]^2$. These are $4m$ vectors of norm at most $\sqrt{2}m$, and consecutive directions differ by at most $\arctan(1/m) \leq 1/m$, the largest gap being between $(m, 0)$ and $(m, 1)$. Let Z be the polygon circumscribed to the unit disk whose edges have the directions of $\pm V_m$; it is centrally symmetric, hence $Z = \sum_{v \in V_m} [-\frac{\ell_v}{2}\hat{v}, \frac{\ell_v}{2}\hat{v}]$ with $\hat{v} = v/\|v\|$, and $1 \leq h_Z \leq \sec(\frac{1}{2m})$. For $t \geq 0$ set $n_v(t) = \lfloor t\ell_v/\|v\| \rfloor$ and

$$D(t) = \bigoplus_{v \in V_m} n_v(t) \{0, v\},$$

the Minkowski sum of $n_v(t)$ copies of the periodic line $\{0, v\}$ [8]. As t grows, the counts increase. When several counts increase at the same t , we add the corresponding lines one at a time, so the elements form a Minkowski chain of periodic lines starting from $\{0\}$.

Theorem 2. *This chain has mean-radius steps at most $\sqrt{2}m/\pi$, and every element is ε -round with some radius ρ , with $\varepsilon \leq \rho/(8m^2) + 2m^2$.*

Proof. A step adds one periodic line, whose mean radius is $\|v\|/\pi \leq \sqrt{2}m/\pi$. Every element, including those created when ties are broken, satisfies $n_v = t\ell_v/\|v\| - \varphi_v$ for some t and $\varphi_v \in [0, 1]$. The centred support function of $\text{conv } D = \sum_v n_v [0, v]$ is therefore $H = \sum_v n_v |\langle v, u \rangle|/2 = t h_Z - \sum_v \varphi_v |\langle v, u \rangle|/2$. Since

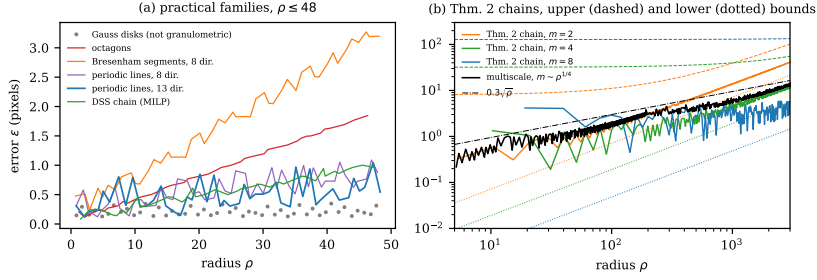


Fig. 2. (a) Error of practical families for $\rho \leq 48$; mean-radius steps are below 2.2. (b) Chains of Theorem 2 for $m = 2, 4, 8$, with the upper bound of Theorem 2 (dashed) and the lower bound of Corollary 1 (dotted), and the multiscale chain against $0.3\sqrt{\rho}$.

$\sum_v \|v\| \leq 4\sqrt{2}m^2$ and $\sec y - 1 \leq y^2$ for $y \leq \frac{1}{2}$, we get $t - 2\sqrt{2}m^2 \leq H \leq t + t/(4m^2)$. The disk of radius $\rho = (\max H + \min H)/2 \geq t - 2\sqrt{2}m^2$ is thus within $\varepsilon \leq t/(8m^2) + \sqrt{2}m^2 \leq \rho/(8m^2) + \sqrt{2}m^2 + \frac{\sqrt{2}}{4} \leq \rho/(8m^2) + 2m^2$. $\square$

With $\delta = \sqrt{2}m/\pi$, Theorem 2 gives $\varepsilon \leq \rho/(4\pi^2\delta^2) + \pi^2\delta^2$, while Corollary 1 gives $\varepsilon \geq (1 - o(1))\rho/(16\pi^2\delta^2)$. For $\rho \geq 4\pi^4\delta^4$, where the constant term is dominated, the best achievable error is therefore $\Theta(\rho/\delta^2)$, with constants within a factor of 8, and of 4 asymptotically. We can also concatenate chains with $m_j = 2^j$, each run up to $t_j \propto 16^j$ on top of the frozen previous ones; anisotropies and radii add. The result is a single chain with steps $O(\rho^{1/4})$ and error $O(\sqrt{\rho})$, which is optimal for steps $O(\rho^{1/4})$ by Corollary 1. Numerically (Fig. 2b), the fixed- m chains have slope about $1/(18m^2)$, between the bounds $1/(32m^2)$ and $1/(8m^2)$, and the multiscale chain satisfies $\varepsilon \leq 0.28\sqrt{\rho}$ for $20 \leq \rho \leq 5 \cdot 10^4$.

Like all our bounds, Theorem 2 concerns convex hulls. Minkowski sums of periodic lines can leave isolated holes, as Jones and Soille's decompositions also do [8]. Adding the 2×2 square to the chain reduces them to 0.2–2% of the hull's lattice points in our tests, and holes do not affect the granulometric property.

Higher dimensions. In $\mathbb{Z}^n$ the directions are not ordered, but a measure argument replaces Lemma 1. We state the result with unspecified constants; its exponent is not sharp, since for $n = 2$ it gives ℓ^{-4} instead of ℓ^{-2} .

Theorem 3. *Let (D_k) be a Minkowski chain in $\mathbb{Z}^n$, $n \geq 2$, whose increments have diameter at most $\ell \geq 1$, and let D_0 and D_K be ε_0 - and ε_K -round (with respect to Euclidean balls). There are constants $b_n > 0$ such that (3) holds with c replaced by $1 - \cos \alpha_n$, where $\sin \alpha_n = 1/(4b_n(2\ell + 1)^n)$. Hence $\varepsilon_K \geq c_n(\rho_K - \rho_0 - \varepsilon_0)\ell^{-2n} - \varepsilon_0$. If the mean-radius steps are at most δ , then $\ell \leq 2\delta/\mu_n$ with $\mu_n = \Gamma(\frac{n}{2})/(\sqrt{\pi}\Gamma(\frac{n+1}{2}))$, and bounded steps again force linear growth.*

Proof (sketch). The directions u split into regions in each of which a single vertex of $Q = \frac{1}{2}(P - P)$ attains $h_Q(u)$ (the normal fan of Q). The boundaries between

Table 2. Families for $\rho \leq 48$: largest mean-radius step δ , maximum and mean error, and the largest radius reached with $\max \varepsilon \leq 1$. The sup-closure uses several elements per size.

family	gran.	max step δ	max ε	mean ε	ρ at $\varepsilon \leq 1$
Gauss disks (disk)	no	1.19	0.35	0.21	–
sup-closure of Gauss disks	yes	1.19	0.35	0.21	≥ 48
octagons (neighbourhood seq.)	yes	2.17	1.85	0.98	25
Bresenham/DSS, 8 directions	yes	1.48	3.27	1.75	12
periodic lines, 8 dir., LP weights	yes	1.42	1.09	0.60	41
periodic lines, 13 dir., LP weights	yes	1.46	1.04	0.48	46
DSS, $\ v\ \leq 6$, MILP	yes	2.09	1.00	0.57	44

these regions lie in the hyperplanes $e^\perp$ for the edges e of the $\text{conv } C_k$, i.e. lattice vectors with $\|e\| \leq \ell$. There are at most $(2\ell + 1)^n$ of them, and each band $\{u : |\langle u, e/\|e\| \rangle| < s\}$ of the sphere has measure at most $b_n s$. So some u_0 satisfies $|\langle u_0, e \rangle| \geq 2 \sin \alpha_n \|e\|$ for all e , and the geodesic cap of radius α_n around u_0 meets no boundary. On this cap $h_Q(u) = \langle q, u \rangle$ is linear. Restricting it to the great circle through u_0 and q gives a variation of at least $|q|(1 - \cos \alpha_n)$, and Steps 3 and 4 are as before. Finally, a segment of length L has mean width $\mu_n L$, and mean width is monotone under inclusion. $\square$

6 Experiments

Families. We compare Gauss disks (**disk**), their sup-closure (Proposition 1), and five single-element families with mean-radius steps between 1.2 and 2.2, i.e. about one element per integer radius. These are octagons; Bresenham/DSS segments as in [1]; periodic lines as in [8] in 8 and in 13 directions (the D_4 -images of $(1, 0)$, $(1, 1)$, $(2, 1)$ and, for 13, also $(3, 1)$); and a chain of individual DSS computed by mixed-integer programming (HiGHS [6]), which minimises the maximal error for $r \leq 48$. For the segment libraries, we take the optimal weights λ^* of Corollary 3 and move towards the multiplicities $\text{round}(t\lambda^*)$ one elementary set at a time, choosing at each step the set that keeps the element roundest. These chains contain the 2×2 square, and all their elements were verified to be hole-free.

Table 2 and Fig. 2(a) summarise roundness. Among single-element families, the optimised chains halve the error of octagons and reach a third of that of Bresenham decompositions, but they remain three times less round than the disks used by the sup-closure. At these radii Corollary 1 only forces 0.06–0.14 for the steps in the table. The lower bound is far from binding for small disks; its force lies in the growth rate (Fig. 2b). As Theorem 1 predicts through ℓ , larger steps help: with radius steps up to 4.1, the 13-direction chain reaches 0.41. With real multiplicities, a linear program finds monotone DSS families with error at most 0.43 up to radius 2000 (numerically; not proven), which points to integrality, used in Step 1 of the proof, as the obstruction.

Table 3. Top: rotation sensitivity $S (\times 10^3)$ of size distributions; in parentheses, the number of increases when nonzero. Bottom: per-object sieve size on `coins` (21 objects, inscribed radius 18–30 pixels; one merged object larger than the tested radii is excluded), error and rotation standard deviation in % of the inscribed radius.

	Gauss	sup-cl.	octag.	DSS 8	per. 8	per. 13	MILP
squares (synthetic)	4.0 (17)	4.0	30.2	13.3	11.8	9.5	17.6
nuclei (microscopy)	1.7	1.7	1.9	1.9	2.0	1.6	1.4
coins	5.0	5.0	5.0	6.4	4.2	5.0	6.2
gravel	1.4	1.4	1.3	1.6	1.1	1.6	1.2
coins: size error (%)	2.8	2.8	4.8	5.7	4.6	4.1	3.8
coins: rotation std (%)	0.65	0.65	1.13	1.20	1.06	0.54	1.00

Size distributions under rotation. For each scene X , rotated by every multiple of 5° up to 45° , we compute $F_\theta(r) = |\gamma_r(X_\theta)|/|X_\theta|$, implementing openings by chain elements as cascades of erosions and dilations by their elementary sets. We report the rotation sensitivity $S = \text{mean}_r \text{std}_\theta F_\theta(r)$ and the number of increases $F_\theta(r+1) > F_\theta(r)$, which are impossible for a granulometry. The scenes are nine squares of sides 17 to 81 pixels, rasterised exactly after rotation ($r \leq 40$), and three real images from `scikit-image`: fluorescence microscopy of nuclei (`human_mitosis`, $r \leq 15$), `coins` ($r \leq 30$), and a material texture (`gravel`, $r \leq 20$). The real images are smoothed, rotated with bilinear interpolation, and thresholded at the Otsu level of the unrotated image.

Per-object sieve size. Sieving assigns to each object O the size $s(O) = \max\{r : \gamma_r(X) \cap O \neq \emptyset\}$ of the largest element that fits inside it, which is the same for Gauss disks and their sup-closure. We compare $s(O)$ with the Euclidean inscribed radius, the maximum of the distance transform in O . We report the mean relative error and the mean standard deviation of $s(O)$ under rotation, relative to the inscribed radius, on objects matched by centroid.

Results. Table 3 and Fig. 3 confirm the picture from the theory.

- *The sup-closure is the best choice.* On every scene it has exactly the rotation sensitivity of Gauss disks, it never increases, and it removes the 17 increases that single Gauss-disk openings produce on the synthetic scene; on the random images of Sect. 3, it removes all increases (0 of the 400 images, against 253 of 400 for single openings).
- *Single-element families pay for their polygonal shape.* On the synthetic scene, octagons are 7.5 times more orientation dependent than the disks, and the 13-direction periodic chain reduces this to 2.4 times.
- *The cost shows on large objects.* Per object, on `coins` (inscribed radii 18–30 pixels), octagons measure sizes with 4.8% error and 1.1% rotation scatter, against 2.8% and 0.65% for the disks. The 13-direction chain has the smallest scatter (0.54%) and a 4.1% error.

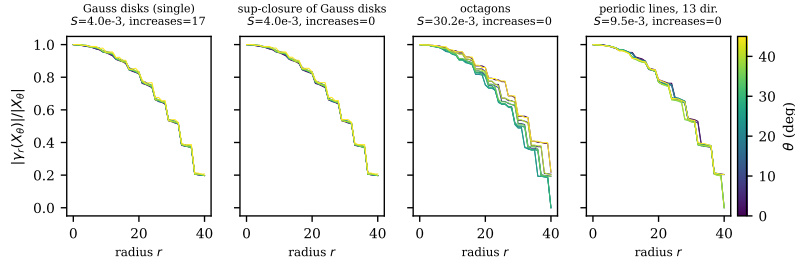


Fig. 3. Size distributions of the rotated synthetic scene, one curve per angle. Single openings by Gauss disks are nearly isotropic but non-monotone. Their sup-closure removes the increases and keeps the isotropy. Octagons are monotone but orientation dependent, and the 13-direction periodic chain reduces this dependence by a factor of three.

- *It does not show on small objects or on whole images.* On nuclei and gravel, with inscribed radii of 3–10 pixels, discretisation dominates: every family errs by 13–24% and none is consistently best. On whole size distributions of the real images, all families agree within the effect of interpolation and thresholding.

The anisotropy of single-element families matters most for large or polygonal objects. The failure of single disk openings is a correctness issue on any image.

Recommendations.

- To compute size distributions or pattern spectra with disks, do not open by each library disk separately. Use the sup-closure of Proposition 1: open by every radius from the largest down and accumulate the unions. This costs nothing extra and yields a correct and round granulometry.
- If a single structuring element per size is required, e.g. to implement each opening as a fixed cascade of two-point erosions and dilations whose cost grows linearly rather than quadratically with the radius, use a Minkowski chain with LP weights. Up to radius about 50, the 13-direction periodic-line chain has error at most 1.04 pixels. For larger radii, a library with slope κ loses $\frac{\kappa}{1+\kappa}\rho + O(1)$ (Corollary 3), and the best library grows with the range of radii (Fig. 2b).

Our code and the computed chains are available¹.

7 Conclusion

The most direct way of measuring size distributions with disks, opening by each digital disk of a library, is not a granulometry, and its size distributions can

¹ Repository link withheld for review.

increase. The fix is simple: take the sup-closure, which needs several structuring elements per size and is correct, round and no more expensive. Single-element families cannot do as well, and we quantified exactly why. With mean-radius steps δ , their error is $\Theta(\rho/\delta^2)$, it is $\Theta(\sqrt{\rho})$ when $\delta \sim \rho^{1/4}$, and bounded steps force linear growth in every dimension. Three questions remain open: the optimal rate when only the fitted radius has bounded steps, between $\rho^{1/3}$ and $\rho^{1/2}$; whether single-element granulometries that are not Minkowski chains, built from unions of translates as in [11], can escape the bound; and matching upper bounds in $\mathbb{Z}^3$.

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